Bi-Weekly Quiz schedule: 19/1, 2/2, 16/2, 23/3, 6/4, 20/4

25/2 midterm
30/4 endterm

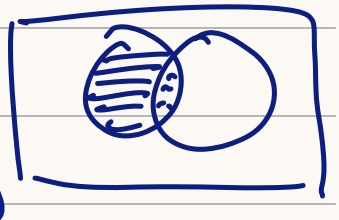
Programming assignments: 15/3, 24/4

Dec 3 :

Derived properties

① $P(A \setminus B) \leq P(A)$

$$A = (A \setminus B) \cup (A \cap B)$$



(additivity) $P(A) = P(A \setminus B) + \underbrace{P(A \cap B)}_{\geq 0}$.

$$\Rightarrow \quad P(A) \geq P(A \setminus B).$$

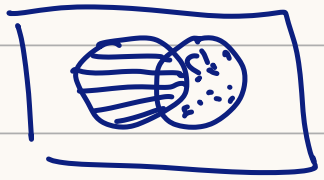
② $A \subseteq B \qquad P(A) \leq P(B)$.

$$B = B \cap \Omega$$
$$= B \cap (A \uplus A^c). \quad \longrightarrow \text{union of disjoint sets}$$
$$= \underbrace{(B \cap A)}_{= A} \cup \underbrace{(B \cap A^c)}_{= B \setminus A}.$$

(additivity) $\Rightarrow \quad P(B) = P(A) + P(B \setminus A)$

(non-negativity) $\Rightarrow \quad P(B) \geq P(A)$

③ Union bound $\quad P(A \cup B) \leq P(A) + P(B)$.



$$A \cup B = A \uplus B \setminus A$$
$$P(A \cup B) = P(A) + P(B \setminus A).$$
$$\leq P(A) + P(B).$$

General union bound

$$P\left(\bigcup_{i=1}^{n} A_n\right) \leq \sum_{i=1}^{n} P(A_i)$$

Sample space. (Ω)

Examples :

finite

① $\Omega = \{1, 2, 3, 4, 5, 6\}$ one die roll.

② $\Omega = [6] \times [6]$ two dice roll

Countable infinite

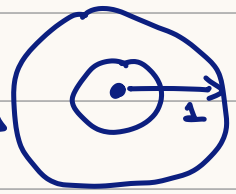
③ $\Omega = \{H, TH, TTH, \dots\}$.

uncountable infinite

④ $\Omega = [0, 10]$

⑤ $\Omega = [0, 1]$.

⑥ $\Omega = \{(x, y) \mid x^2 + y^2 \leq 1\}$



Event Space Examples :

① $\Omega = \{1, 2, 3, \dots 6\}$

ⓐ $\mathcal{F}$ all subsets of Ω i.e., $\mathcal{F} = 2^{\Omega}$.

ⓑ $\mathcal{F} = \{\phi, \Omega, \{1, 3, 5\}, \{2, 4, 6\}\}$

② Ω let $A \subseteq \Omega$.

ⓐ $\mathcal{F} = \{\phi, \Omega, A, A^c\}$.

Excercise

ⓑ $\mathcal{F} = \{\phi, \Omega, A, B, A^c, B^c, A \cap B, A \cup B, A^c \cap B, A^c \cup B, \dots\}$.

$\Downarrow$

$\mathcal{F}$ is generated by sets A, B.

③ $\Omega = \mathbb{R}$ (real numbers).

Excercise.

$\mathcal{F}$ generated by $\{(-\infty, x] \mid x \in \mathbb{R}\}$

Open intervals, closed intervals, points are elements in $\mathcal{F}$.

(a, b), $[a, b]$, $[a, b)$, $(a, b]$, $\{x\}$ are elements in $\mathcal{F}$.

Probability function

Examples:

① $\Omega = \{1, 2, .. 6\}$

$\mathcal{F} = 2^{\Omega}$ (all subsets of Ω)

ⓐ $P(\{i\}) = \frac{1}{6} \quad \forall i \in \Omega.$

$P(A) = \frac{|A|}{6}$

$A \in \mathcal{F}$ and $A \subseteq \Omega$

$P(\{1,2\}) = P(\{1\}) + P(\{2\})$

$= \frac{1}{6} + \frac{1}{6} = \frac{2}{6}.$

$P(\Omega) = \frac{|\Omega|}{6} = 1.$

ⓑ Suppose the die is not fair.

$P(\{6\}) = \frac{1}{2}, \quad P(\{i\}) = x$

$\forall i \neq 6.$

$1 = P(\Omega) = \sum_{i=1}^{6} P(\{i\}) = P(\{6\}) + 5x$

$= \frac{1}{2} + 5x$

$5x = \frac{1}{2} \Rightarrow x = \frac{1}{10}.$

② $\Omega = \{H, TH, TTH, ...\}$

Suppose. $P(\{\underbrace{TT\cdots T}_{n-1} H\}) = x \quad \forall n.$

equal probability assignment is not possible $\Leftarrow \sum_{n=1}^{\infty} x = 1.$

$P(\{\underbrace{TT\cdots T}_{n-1} H\}) = x_n.$

$$\Rightarrow \quad \sum_{n=1}^{\infty} x_n = 1.$$

geometric progression. $\frac{1}{2}, \frac{1}{2^2}, \frac{1}{2^3}, \ldots$

$x_n = \frac{1}{2^n}$ is a possibility.

---

③ $\Omega = [0,1]$.

define $P((a,b)) = b-a$.

$$P(\Omega) = 1$$

$$P((0,1)) = 1.$$

$P(\Omega) = P((0,1)) + P(\{0,1\}) \Rightarrow P(\{0,1\}) = 0$

$\Rightarrow P(\{0\}), P(\{1\}) = 0.$

<u>Claim:</u> $P(\{x\}) = 0$. for $\forall x \in [0,1]$.

<u>Proof:</u> $P((0,1) \setminus \{x\}) = P((0,x) \cup (x,1))$.

from additivity $= P((0,x)) + P((x,1))$.

from the defn. $\leftarrow = x + 1-x$

$= 1 \Rightarrow P(\{x\})$

$= P((0,1)) - P((0,1) \setminus \{x\})$

$= 0$

<u>Claim:</u> $P(\mathbb{Q} \cap [0,1]) = 0$

<u>Proof:</u> Since $\mathbb{Q}$ is countable

$\mathbb{Q} \cap [0,1]$ is countable.

$$\mathbb{Q} \cap [0,1] = \{x_1, x_2, x_3, \ldots\}.$$

$$P(\mathbb{Q} \cap [0,1]) = P\left(\bigcup_{n=1}^{\infty} \{x_n\}\right)$$

$$= \sum_{n=1}^{\infty} P(\{x_n\}).$$

$$= 0.$$

<u>Countinuity property of probability measure.</u>

$$P\left(\bigcup_{i=1}^{\infty} A_i\right) = \lim_{n\to\infty} P\left(\bigcup_{i=1}^{n} A_i\right).$$

<u>Proof</u>: $A_1, A_2, A_3, \ldots$

We'll construct $B_1, B_2, B_3, \ldots$ that are disjoint

st $\bigcup_{i=1}^{n} B_i = \bigcup_{i=1}^{n} A_i$

$$B_1 = A_1$$

$$B_2 = A_2 \setminus A_1$$

$$B_3 = A_3 \setminus (A_1 \cup A_2).$$

$$\vdots$$

$$B_i = A_i \setminus (A_1 \cup A_2 \cup \cdots A_{i-1})$$

$$\boxed{\bigcup_{i=1}^{n} B_i = \bigcup_{i=1}^{n} A_i}$$ excercise.

$$\bigcup_{i=1}^{\infty} B_i = \bigcup_{i=1}^{\infty} A_i .$$

clearly $\bigcup_{i=1}^{\infty} B_i \subseteq \bigcup_{i=1}^{\infty} A_i$

as $B_i \subseteq A_i \ \forall i$

$x \in \bigcup_{i=1}^{\infty} A_i \implies x \in A_i$ for some i

$\implies x \in \underbrace{\bigcup_{j=1}^{i} A_j = \bigcup_{j=1}^{i} B_j}_{\text{from excercise statement}}$

$\implies x \in B_{\hat{i}}$ for some $\hat{i}$.

$\implies x \in \bigcup_{\hat{i}=1}^{\infty} B_{\hat{i}}$

$\implies \bigcup_{i=1}^{\infty} A_i \subseteq \bigcup_{i=1}^{\infty} B_i$

$$P\left(\bigcup_{i=1}^{\infty} A_i\right) = P\left(\bigcup_{i=1}^{\infty} B_i\right)$$

additivity axiom $= \sum_{i=1}^{\infty} P(B_i)$

$$= \lim_{n\to\infty} \sum_{i=1}^{n} P(B_i)$$

$$= \lim_{n\to\infty} P\left(\bigcup_{i=1}^{n} B_i\right)$$

(from excercise) $= \lim_{n\to\infty} P\left(\bigcup_{i=1}^{n} A_i\right)$